End-to-End Target Liveness Detection via mmWave Radar and Vision Fusion for Autonomous Vehicles

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The successful operation of autonomous vehicles hinges on their ability to accurately identify objects in their vicinity, particularly living targets such as bikers and pedestrians. However, visual interference inherent in real-world environments, such as omnipresent billboards, poses substantial challenges to extant vision-based detection technologies. These visual interference exhibit similar visual attributes to living targets, leading to erroneous identification. We address this problem by harnessing the capabilities of mmWave radar, a vital sensor in autonomous vehicles, in combination with vision technology, thereby contributing a unique solution for liveness target detection. We propose a methodology that extracts features from the mmWave radar signal to achieve end-to-end liveness target detection by integrating the mmWave radar and vision technology. This proposed methodology is implemented and evaluated on the commodity mmWave radar IWR6843ISK-ODS and vision sensor Logitech camera. Our extensive evaluation reveals that the proposed method accomplishes liveness target detection with a mean average precision (mAP) of 98.1%, surpassing the performance of existing studies.

CCS Concepts: • **Hardware → Sensor devices and platforms**; **Sensor applications and deployments**.

Additional Key Words and Phrases: target liveness detection, mmWave radar

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1 INTRODUCTION

Autonomous vehicles must have accurate and robust sensing abilities in complex real-world road environments. Liveness target detection is one critical sensing ability. On the one hand, misidentifying the living targets might lead to serious car accidents (e.g., a Tesla autopilot caused a fatal motorcycle accident in 2022 [4]). On the other hand, a false alarm also causes trouble. For example, roadside billboards with human portraits will bring serious visual interference to the perception and decision of autonomous vehicles, which could lead to erroneous control strategies such as sudden braking, unexpected changes in direction, and sudden speeds up [1]. In the future full autonomous driving (e.g., L5), it is very significant for vehicles to prevent these mistakes and distinguish the living targets from visual interference accurately.

In the real-world environments of autonomous driving, a diverse range of visual interferences, such as LED billboards on vehicles, often exist, leading to the inaccuracy of methods solely based on visual perception. Despite this, the most common sensors in autonomous driving vehicles are mmWave radar and camera [7, 17]. Cameras offer abundant visual data for identifying road objects but struggle to differentiate between visually alike targets and often fail to discern the liveness of dynamic visual interference (e.g., a video of a biker) displayed on LED billboard advertisements. Conversely, the mmWave radar utilizes the reflection of millimeter wave signals for sensing unaffected by visual interference and robust under harsh sensing environments, e.g., dark, foggy, and rainy conditions. Nonetheless, mmWave radar possesses its own constraints. The point clouds produced by mmWave radar tend to be sparse, noisy and have a significantly lower angular resolution compared to camera imagery [35], which is insufficient for accurate target liveness detection. Moreover, the performance of radar liveness target detection depends on the target's orientation, which is difficult to obtain from the point cloud due to its sparsity.

Motivated by the necessity for liveness target detection and the potential of multi-modal fusion, this work presents the first end-to-end methodology for liveness target detection harnessing the strengths of mmWave radar and vision technologies. The key idea underpinning our design is to leverage the inherent complementarity between radar and camera technologies. In particular, radar provides liveness-related information that is impervious to visual interference, albeit with some sparsity and noise [37]. In contrast, vision data, while susceptible to visual interference, furnishes detailed visual features that enable precise target segmentation.

In light of the key insight, we propose a multi-modal data fusion machine learning scheme that collaborates the mmWave radar with vision sensors for target liveness detection. We exploit the unique features in the mmWave radar and the vision data. In the mmWave radar data, the Radar Cross Section (RCS) is extracted as the distinguishable characteristic for liveness detection. The RCS, influenced by numerous target features such as shape, size, and material, serves as a comprehensive metric. Utilizing the off-the-shelf mmWave radar device, we measure the RCS of the living target and visual interference. The result shows that the RCS of the living target significantly differs from the visual interference. Moreover, the RCS values are stable under the detection ranges of mmWave radar, which means as long as the targets are within the field of vision, the results are not affected by the distance from the object to the radar.

Meanwhile, we obtain detailed visual features from the visual data to estimate the target's posture, offering essential angle-of-view insights to counteract the radar signal's sensitivity to the target's posture. Finally, mmWave radar and visual features are collaboratively utilized to identify the true living targets and prevent visual interference. To achieve this, we extend MediaPipe [20] (a cross-platform, customizable machine-learning methodology for target detection) to support mmWave radar and vision multi-modal data fusion. This fusion of technologies leads to a refined liveness target detection process, effectively handling scenarios where a human target coexists with metal, such as bikers.

Consequently, our contribution transcends individual advancements in RCS and vision technologies, promoting a novel collaboration for superior liveness target detection. The approach we've established forms a solid foundation for future exploration of additional features that could further enhance detection robustness and accuracy.

This paper has the following contributions:

- This paper presents the first end-to-end liveness target detection methodology utilizing the mmWave radar and vision fusion, which enhances the robustness of autonomous vehicles.
- This paper presents a novel design that extracts the RCS feature from mmWave radar signal and utilizes a multi-modal data fusion machine learning method for mmWave radar and vision data fusion. The proposed methodology utilizes the complementary information in the mmWave radar and vision data for robust liveness target detection, thus avoiding visual interference.
- We implement and evaluate the performance of the proposed design on the off-the-shelf mmWave radar board IWR6843ISK-ODS and vision sensor Logitech camera by experimenting with the system in different scenarios. The results demonstrate that the proposed method achieves the liveness target detection with an mAP of 98.1%.

2 MOTIVATION

2.1 The Necessity of Liveness Target Detection in Autonomous Vehicles

Accurately distinguishing living targets from visual interference is important for autonomous vehicles to make correct and safe control. In the real-world road environment, the living target looks similar to the visual interference. For example, autonomous vehicles still do not have the capacity to fully distinguish a real biker from the bikers in advertisements on billboards along the road utilizing visual information only. These mistakes make autonomous vehicles execute incorrect actions and lead to serious accidents. On the other hand, being unable to distinguish living targets from visual interference also results in serious issues. For instance, Tesla's autopilot was involved in a third fatal motorcycle crash in 2022, raising questions about the driver-assist system's ability to operate safely [4]. A robust liveness target detection is needed to improve the safety of autonomous vehicles. Specifically, when the autonomous vehicle faces living targets and visual interference, our design aims to detect living and exclude visual interference (e.g., billboard).

2.2 Limitation of Existing Solutions

In recent years, research progress has been made in the field of target detection based on mmWave radar and vision data. The traditional target detection approaches, e.g., CRUW [39], CARRADA [23], nuScenes [6], LiveNet [26], and RadarScenes [31], are based on the position or intensity of the body to achieve object detection and recognition. In general, there are two kinds of commonly utilized methodologies for achieving liveness target detection. One is based on the contour feature of the face to realize liveness target detection [26, 39], but these methods are only utilized in the case of close faces, so they are not suitable for automatic driving scenes. The other method is based on the contour features of the human body to realize the detection of living targets [25, 27], and the implementation effect of this method in the autonomous driving scene is relatively much better.

However, these methods fail to distinguish living targets from visual images, e.g., recognizing a billboard with a human portrait as a real person. This is because living targets and visual interference may have similar visual characteristics, so visual information alone is not enough for liveness target detection. In this work, we propose a new design that takes advantage of mmWave characteristics to make the liveness target detectors robust to the disturbance of visual images.

2.3 Opportunities of mmWave Radar and Vision Fusion for Liveness Target Detection

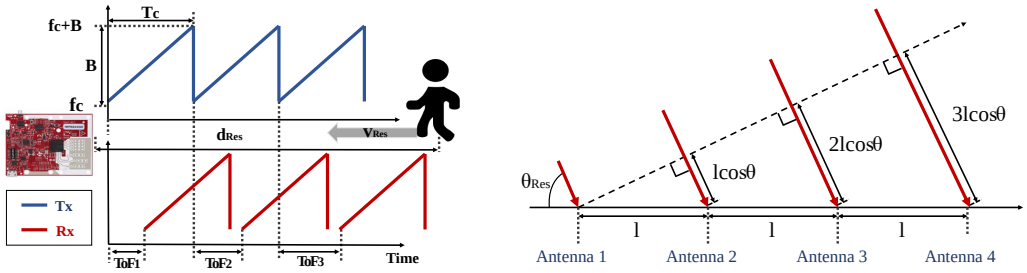
Most autonomous vehicles have vision sensors, i.e., cameras and mmWave radars that are fused for liveness target detection besides visual and wireless sensing. The camera provides the capacities of target segmentation and posture recognition from visual data. In contrast, the mmWave radar brings the opportunity of liveness target detection to distinguish living targets from visual interference. Moreover, the radar cross-section (RCS) feature extracted from the mmWave radar signal is stable with visual interference, such as billboards, which solves the limitations of visual sensors. In conclusion, the mmWave radar and vision sensors have complementary features for robust liveness target detection.

3 BACKGROUND

This section introduces the primer knowledge of the mmWave radar and machine learning methodologies utilized in this work.

3.1 mmWave Radar Principles

The fundamental principle of mmWave radars is straightforward. A mmWave signal is generated and transmitted from the radar toward a target that scatters the transmitted mmWave signal. Some scattered waves are redirected back toward and intercepted by the receiving antenna. The received wave contains information from different dimensions, e.g., time, space, and frequency. This information is extracted and supports the mmWave radars' fundamental range, velocity, and angle measurement functions. This subsection introduces the principle for those functions in an easy-to-understand way. The key idea behind frequency-modulated continuous-wave radar is to measure the Time-of-Flight (ToF) of signal from antennas by comparing it with its delayed signal reflected by the target. As shown in Fig. 1, the range measurement is done by multiplying half of the ToF with the signal speed. The velocity measurement is done by detecting the cross-chirps ToF differences, as shown in Fig. 1(a). Moreover, the angle measurement is done by estimating the cross-antenna ToF differences, as illustrated in Fig. 1(b).



(a) Radar Multiple Chirps (b) Antennas Array

Fig. 1. (1) Compute the range information from one ToF; (2) Compute the velocity information from over-chirps ToF differences; (3) Compute the angle information from over-antenna ToF differences.

Range Measurement. Range measurement is done by measuring the propagation time of the reflected wave and multiplying it by the wave propagation speed. Specifically, the mmWave radars transmit frequency-modulated continuous wave radio to measure the range. After the antenna sends the signal, it will be blocked by objects in the transmission path and reflected. The receiving antenna of mmWave radar will capture the transmitting linear frequency modulation pulse. A mixer inside the radar combines the sent signal with the captured reflected signal to produce the intermediate

frequency signal. Then, the fast Fourier transform operation is performed on the intermediate frequency signal to separate different frequency components and thus get the distance between each object and the radar denoted as $d = \frac{cfT_c}{2B}$. Based on the Fourier transform theory, the range measurement resolution is limited as $d_{Res} = \frac{c}{2B}$, which depends on the wave frequency bandwidth. For example, a bandwidth of 4GHz gives a range resolution of 3.75cm.

Velocity Measurement. The movement of the target results in a frequency shift of the received wave determined as the Doppler shift. The Doppler shift value is utilized for velocity measurement. The principle of velocity measurement is finding the distance change amount in a short period, i.e., T_c. The FMCW technology continuously emits a set of N equally spaced linear frequency modulation pulses, called a frame, to obtain multiple phases after the distance fast Fourier transform and then perform the Doppler fast Fourier transform called slow time fast Fourier transform, on multiple phases. The phase value w of different objects are obtained, and thus obtain the amount of change in distance, i.e., $\Delta d = \frac{\lambda \Delta \phi}{4\pi d}$, Finally, the velocity $v = \frac{\lambda w}{4\pi T_c}$ is obtained. Also, according to Fourier transforms theory, the velocity resolution is denoted as $v_{Res} = \frac{\lambda}{2T_f}$ where $T_f = NT_c$, which depends on the length of the observation window. Larger window size leads to higher velocity measurement resolution.

Angle Measurement. The Angle-of-Arrival (AoA) measurement represents the receiving wave direction of the antenna array. The principle of AoA measurement is that the same object is at different distances to different receiving antennas, and the distance difference has a trigonometric relationship with the AoA. So as long as this distance difference is calculated, the AoA is calculated based on the geometric relationship between it and the distance l between antennas. Similar to the Doppler fast Fourier transform, it is necessary to perform a fast angular Fourier transform to distinguish multiple objects at the same distance with the same velocity to separate the components of different objects. After obtaining $\Delta\phi$, the angle θ is calculated by: $\theta = sin^{-1}(\frac{\lambda \Delta \phi}{2\pi l})$. The angle measurement resolution is calculated as $\theta_{Res} = \frac{\lambda}{Ndcos(\theta)}$. Angle measurement resolution is related to the number of the antenna array. The more receiving antennas, the better the angular resolution.

3.2 Visual-based Machine Learning Principles

Visual methods based on machine learning have been widely utilized in 3D target detection and liveness recognition. However, due to the need for more data and the diversity of the appearance of the recognized objects, it is a challenging research topic to detect 3D objects. Therefore, we choose the state-of-the-art 3D target recognition framework MediaPipe[20] and related algorithms to implement the 3D vision-based target recognition part of our model.

Target Segmentation. Under the development framework of MediaPipe, the model of a single-stage pipeline is utilized to achieve target segmentation. The backbone of the model is an encoder-decoder framework based on the MobileNetv2 model, and a multi-task learning method is utilized to realize object detection and object shape frame prediction. The 2D projections of eight boundary points of the object shape frame are estimated by regression. Finally, a mature pose estimation algorithm EPnP is utilized to recover the 3D bounding box of the object to realize the detection of the object and the estimation of the shape and pose.

Posture Recognition. The MediaPipe framework also implements human pose detection based on a two-stage detector-tracker ML pipeline. The tracker first utilizes a detector to locate the person/pose Region of Interest (ROI) within the frame. Frames cropped to the ROI are then utilized as input to predict pose landmarks and segmentation masks within the ROI. Finally, the bounding boxes of different parts of the human body are displayed and rendered in the image, and the human pose is judged through these results.

4 SYSTEM OVERVIEW

This paper introduces an end-to-end liveness target detection system with mmWave radar and vision fusion. The input of this system is signal data from the mmWave radar and vision data from the camera. Fig. 2 presents the overall system architecture. The mmWave signal data from the radar are processed by the RCS calculation methodology (Section 5). The output of the RCS calculation process is the point cloud of the mmWave radar with RCS information. Then, the point clouds are projected into radar pixel images with RCS information. Meanwhile, the vision data from the camera are sent into MediaPipe [20] in the stream. During that, the vision data are divided into frames and processed for target segmentation and posture recognition. The target segmentation process employs advanced image processing techniques to identify potential targets within an image. It segments the image into different regions, each potentially containing a target object. The posture recognition process leverages machine learning techniques to recognize the posture or orientation of the detected objects. It classifies the objects based on their features, providing an additional layer of information that significantly enhances the overall object detection process. The output target segmentation results and the output target posture results are then sent together with the mmWave radar pixel image into two designed fusion models for liveness target detection. The fusion models combine mmWave radar and vision sequence data in a time-series format, marking a significant contribution to our work. They exploit the unique features (e.g., movements) within radar and image sequence data over time, which help to distinguish living targets from dynamic visual interferences such as LED billboards, a situation where conventional state-of-the-art techniques often fall short. Two fusion models are detailed in Section 6. At last, the system outputs bounding box results for the living target to distinguish them from the visual interference.

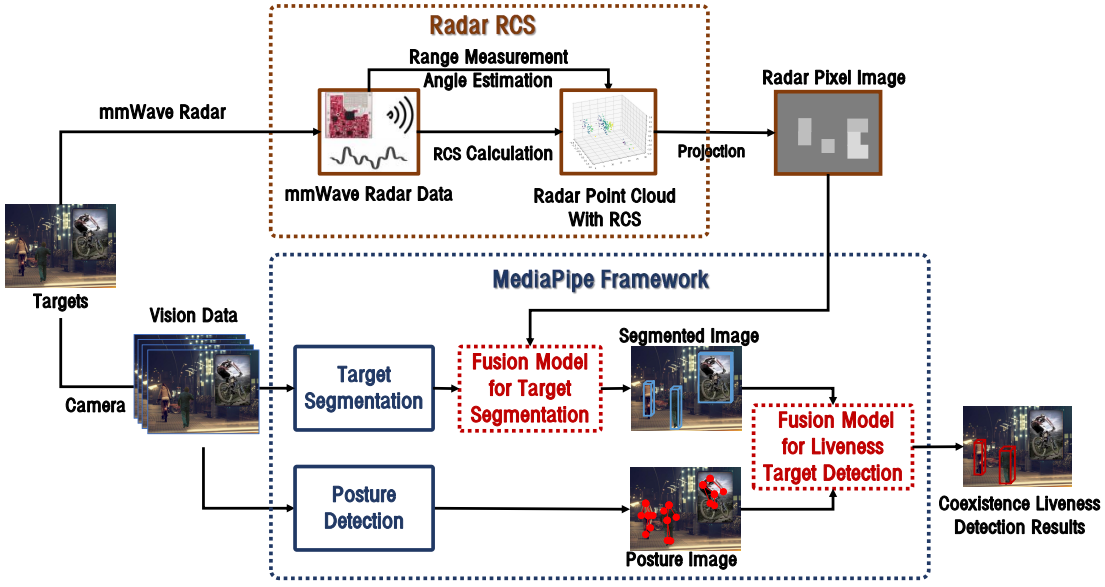


Fig. 2. System architecture.

5 FEASIBILITY OF LIVENESS TARGET DETECTION

This section introduces the radar cross-section (RCS), the crucial feature for mmWave radar liveness target detection. This work first introduces the principle of RCS. Then we illustrate the methodology of RCS calculation in commodity mmWave radars. After that, we provide a detailed study of living targets' RCS measurements.

5.1 Radar Cross Section

Radar cross section (RCS) measures the object's reflection area to radar. A larger RCS indicates that the object is more obvious and easier to be detected by radar [28]. The object's RCS is related to its size, material, and orientation. The RCS features were initially utilized for aircraft and missile detection and distinguishment in the military field. Moreover, we find that RCS features have some unique characteristics which bring opportunities for mmWave and vision fusion liveness target detection. (i) First, the RCS of the living target has a huge difference with its visual interference. (ii) Second, the RCS value is consistent in various sensor-to-target distances. (iii) Third, the RCS of the living target is related to its posture (the angle of view towards radar). Therefore, utilizing the vision data for target posture detection and fusion with mmWave radar data for liveness target detection improves the robustness of the autonomous vehicles' sensing ability.

5.2 RCS Acquisition on Commodity Radar

RCS is a physical quantity that measures the intensity of the echo generated by the target illuminated by the radar wave. We obtain the RCS value of the object reflection to judge the shape characteristics of the target. But off-the-shelf radars (e.g., the TI IWR series) do not provide RCS results directly. Therefore, we need to obtain the other parameters (including the radar parameters) before calculating the RCS value.

In calculating RCS, the most critical and difficult problem is obtaining the SNR, which is the ratio of RX average signal strength and average noise strength. The noise mainly depends on the background noise of the radar circuit. Although the intensity of this noise is relatively stable, it is difficult to measure the noise value on the integrated radar equipment directly. Therefore we do not directly obtain the noise value but consider that the RCS should always be proportional to the RX signal strength when the other parameters are determined. Therefore, we decided to use the corner reflector[10] in the calibration phase to obtain the RX signal strength at different distances d and establish the benchmark database $\mathcal{B}(d)$. From $\mathcal{B}(d)$, we obtain the space Cartesian coordinates (x, y, z), if the distance $L = \sqrt{x^2 + y^2 + z^2}$, then we calculate RX signal intensity P_{r_t} based on the following formula:

$$\sigma_p = \frac{P_{r_t}}{\mathcal{B}L}\sigma_r \tag{1}$$

We obtain the benchmark database $\mathcal{B}(d)$ during the calibration phase. From $\mathcal{B}(d)$, we obtain the signal-to-noise ratio, and then we obtain the RCS based on the following formula:

$$\sigma = \frac{(4\pi)^3 d^4 kTFSNR}{P_t G_{TX} G_{RX} \lambda^2 T_{meas}} \tag{2}$$

Here k is the Boltzmann constant, T is the antenna temperature, F is the noise coefficient of RX, λ is the millimeter-wave wavelength (constant), P_t is the radar output power, G_{TX} is the TX Antenna Gain, G_{RX} is the RX Antenna Gain, T_{meas} is the measurement time, and d is the distance between the target and the millimeter-wave radar. d is computed from the radar output and the metadata. We obtain the RCS value of the target through the above method.

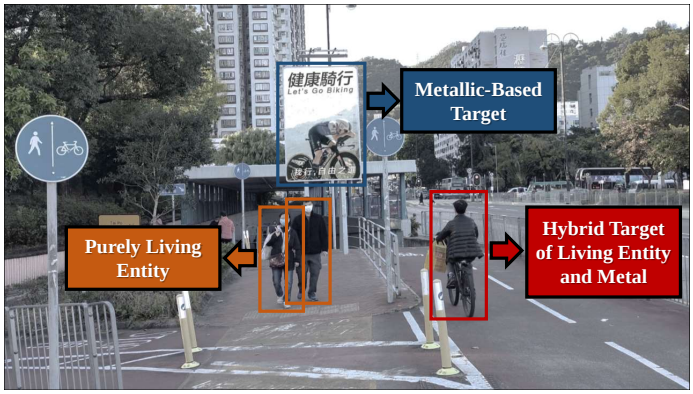


Fig. 3. Experimental scenarios of RCS value differences between living target and metal.

5.3 RCS Value of Living Targets and Visual Interference

We conduct experiments to measure the RCS value of the living targets and the visual interference utilizing the commercial mmWave radar IWR6843-ODS to verify the feasibility of RCS for liveness target detection. As shown in Fig. 3, there are two groups of target objects scenarios. The first scenario contains the metal and living target. The second scenario contains the metal with the hybrid target of living entity and metal advertisement in the same place as the first scenario. The metal is 0.8m × 0.7m in size and has aluminum alloy in the material. We utilize mmWave radar to measure the RCS value of the living targets and the visual interference around the different ranges and different angles of view.

Table 1 illustrates the significant distinction present in the Radar Cross Section (RCS) between living targets and visual interferences. Three broad categories of objects are prevalent on streets: metallic-based targets (e.g., vehicles and billboards), hybrid targets of Living entity and Metal (e.g., bikers), and purely living targets (e.g., pedestrians). RCS is defined as the effective area that intercepts the transmitted radar power and then scatters that power isotropically back to the radar receiver, therefore, the unit of RCS is m^2. Quantitatively, the average RCS for a pedestrian is $3.5m^2$. The RCS value of a Biker is $50m^2$ when observed from the side and $20m^2$ when observed from the front, indicating the influence of metal in the hybrid category. However, metallic objects significantly inflate the RCS values, with the billboard and vehicle documenting figures of $230m^2$ and $500m^2$, respectively. The divergence in these figures emphasizes the potential of liveness target detection within the landscapes of autonomous driving.

Table 1. Experimental scenarios and RCS values of living target and visual interference.

Type	Scenario	Target Object	RCS
Visual Interference	**Metallic-Based Target**		
	1	vehicle	$500m^2$
	1	billboard	$230m^2$
Living Targets	**Hybrid Target of Living Entity and Metal**		
	2	Biker side view	$50m^2$
	2	Biker front view	$20m^2$
	Purely Living Target		
	3	Pedestrian	$3.5m^2$

5.3.1 RCS from Different Distance. We further demonstrate the RCS of the living target and the visual interference when the sensors are at different distances, moving from 5 meters to 30 meters, from the object. As shown in Fig. 4, the

RCS of the hybrid target of living entity and metal from the side, the front view, and the living target is very stable in the range of 47~51m^2, 17~22m^2, and 2~5m^2, respectively. However, the RCS of the metal is in the range of 200~300m^2. Therefore, the RCS value of the living target differs from the visual interference. Moreover, the RCS value of the same target at different distances is generally consistent, which supports the theory that the target's RCS, in the sensing coverage of mmWave radar, is not influenced by the target distance.

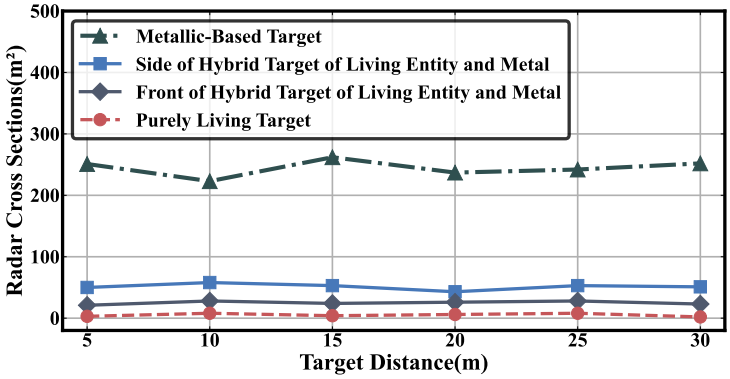


Fig. 4. Experiment result of RCS from different distances.

5.3.2 RCS from Different Angles of View. The signal from the mmWave radar is highly sensitive to an object's posture, indicating a strong correlation between the object's viewing angle and its RCS (Radar Cross Section) value. To quantify this effect, we undertake a series of experiments to assess viewing angle effects on RCS. As illustrated in Fig. 5, we collect radar RCS data for the living target and the visual interference across a range of viewing angles. Our findings highlight the significant variation in the RCS values of both the living target and the visual interference. Moreover, it's notable that despite the RCS value for both the hybrid target of living entity and metal and the metallic-based target varying significantly in alignment with the viewing angle, the hybrid target of living entity and metal distinctly alters based on the target posture. This insight motivates us to further integrate the mmWave radar and vision fusion cues, such as target posture, to fortify the robustness of liveness target detection.

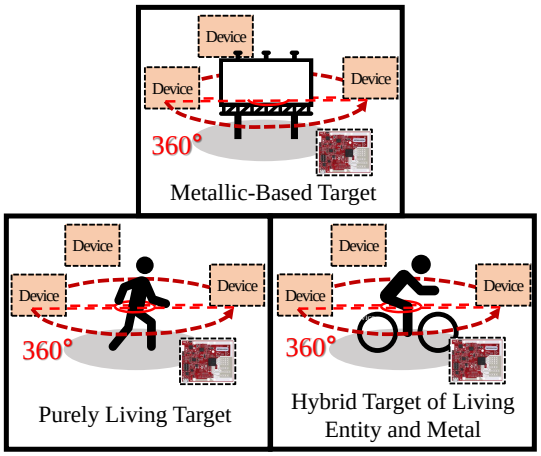


Fig. 5. Experimental scenarios of RCS from different angles of view.

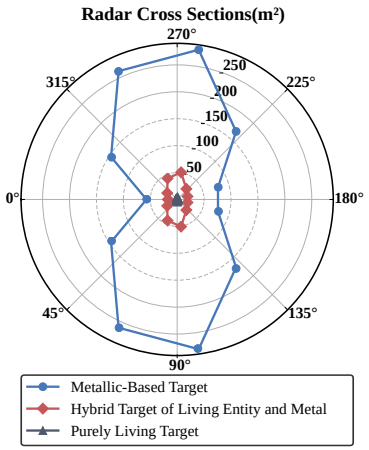


Fig. 6. RCS from different angles of view.

Fig. 6 shows the RCS results from different angles of view. Although RCS is changing related to the angle of view, the change range of the living target and the visual interference is different. The RCS value of the metal rises from $50m^2$ to $300m^2$ when the angle of view changes from 0° to 90°, then the RCS value decrease to $54m^2$ when the angle of view changes from 90° to 180°. Meanwhile, the RCS value of the hybrid target of living entity and metal rises from $17m^2$ to $51m^2$ when the angle of view changes from 0° to 90°, then the RCS value decreases to $20m^2$ when the angle of view changes from 90° to 180°.

5.4 Summary

The experiment results prove that the RCS values of the living target and the visual interference differ significantly. Moreover, the RCS value is related to the angles of view, and vision data contains the target's angle of view information. These bring opportunities for utilizing mmWave radar and vision fusion for liveness target detection.

6 DESIGN

6.1 Design Overview

This section proposes a multi-modal data fusion liveness target detection methodology, which fuses the mmWave radar and vision data to distinguish the living target from visual interference. The common method of utilizing mmWave radar and vision data together is first to process target segmentation by the two-stage image-based detectors (e.g., Faster-RCNN), utilizing the vision data. Then map the mmWave radar data into the corresponding target to distinguish whether it is the living target or the visual interference. However, our experiments demonstrate that the two-stage image-based detectors come into a short while detecting the target at different angles of view and only achieve mAP of 57.5% (detailed results are shown in Section 8).

To achieve robustness in distinguishing the living target from visual interference at different angles of view. This article proposes an end-to-end liveness target detection system with mmWave radar and vision data. The system utilizes mmWave signal data for liveness target detection and vision data for target segmentation and posture detection. During the target segmentation process, this article utilizes machine learning target segmentation methodology for live and streaming media (e.g., MediaPipe [20]). Specifically, visual data streams from the camera are input into the single-stage pipeline in MediaPipe [2]. The pipeline estimates the real-time 3D object poses through the machine learning model. Then the radar pixel images from mmWave radar data are fused into the output of the single-stage pipeline for getting the target segmentation results with the RCS features. Meanwhile, during the posture detection process, the vision data are sent into the two-step detector-tracker pipeline in MediaPipe [3]. The pipeline first locates the target's region of interest in the vision data, and then the tracker subsequently predicts the target's posture within the ROI utilizing the ROI-cropped frame. The output in the target segmentation process and the posture detection process are then sent together into the designed fusion model for getting liveness target detection results. The system outputs bounding boxes around the living targets to distinguish them from visual interference.

6.2 mmWave Radar Data Procession

The RCS values are extracted from the mmWave radar data. This subsection provides the RCS calculation algorithm for obtaining the RCS value of the target object in the 3D space. Because of the sparsity of mmWave radar point cloud, there are noise voxels without useful information for liveness target detection. To solve this problem and extract multi-scale features from mmWave radar data more efficiently, this paper projects the mmWave radar data from the Cartesian

coordinate system to the pixel coordinate system of the vision data by squeezing it on the depth direction (y-axis). As shown in the following equation:

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \frac{1}{y} M_1 M_2 \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \tag{3}$$

Here M_1 is the 3×3 internal parameter matrix of the vision data, M_2 is the 3×4 external parameter matrix of the vision data, (u, v) is the pixel location in the vision data's pixel coordinate system. Because the mmWave radar and vision sensors are fixed at an unchangeable place on the autonomous vehicle, the relative positions between mmWave radar and vision data remain static during the measurement stages. Therefore, M_1 and M_2 are parameters that are known ahead.

After that, the mmWave radar data are processed into a three-channel radar pixel image the same size as the vision data. Then, the target's RCS value is utilized to fill each pixel of the radar pixel image corresponding to the target location on the vision data. As shown in Fig. 7, the left figure is the vision data collected by the vision sensor, and the right is the mmWave radar pixel data calculated from the mmWave radar data. The vision and mmWave radar pixel data are collected and calculated simultaneously.

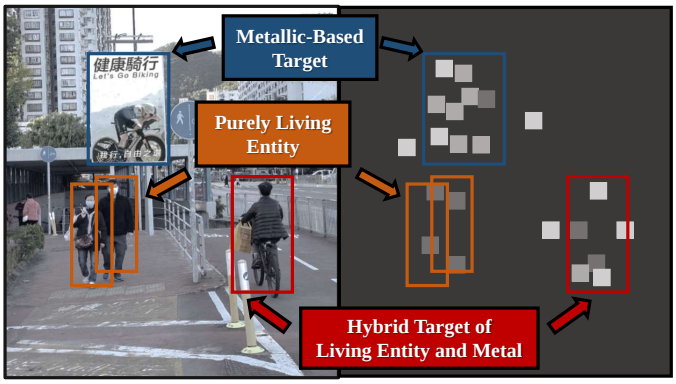


Fig. 7. Image and radar pixel image.

6.3 Multi-modal Feature Extraction

As discussed in Section 6.1, the target objects are located at various angles of view and distances from the autonomous vehicle. The target objects' size in both the vision data and the mmWave radar pixel image decrease when the distance from the target to the sensors increases. To accurately detect the targets and obtain features from sparse mmWave radar data, this paper provides the multi-scale feature extraction methodology for obtaining high-dimensional features from mmWave radar and vision data in different scales. This feature extraction methodology is utilized for the accurate detection of small-scale objects.

As shown in the blue block in Fig. 2, this article utilizes the single-stage pipeline in MediaPipe for target segmentation and posture recognition. The signal-stage pipeline contains the encoder and decoder architecture. The multi-task learning approach is employed for jointly processing target segmentation and posture recognition. This paper utilizes the annotated bounding boxes for the target segmentation task, and the box size is standardized to be the same as the deviation proportion. To obtain the bounding box's final 3D coordinates, we leverage the pose estimation algorithm

[18] for point correspondence, which draws the 3D bounding box of the objects without prior knowledge of the object's shape. Then, this paper further utilizes the multi-modal feature fusion methodology for the liveness target distinguishment.

6.4 Multi-modal Feature Fusion

The mentioned feature extraction module yields the features from the mmWave radar and the vision data. Then, the multi-modal features are utilized for liveness target detection. The common methodology for feature fusion is tensor concatenation. The multi-dimensional linear features' relationships are further extracted utilizing the multi-layer convolution operation. However, since the mmWave radar and the vision data are from different modalities, the convolution network comes short when learning the linear relationship between those two kinds of data. Our preliminary experiments, as shown in Section 8, provide evidence that the single-stage MediaPipe pipeline is better.

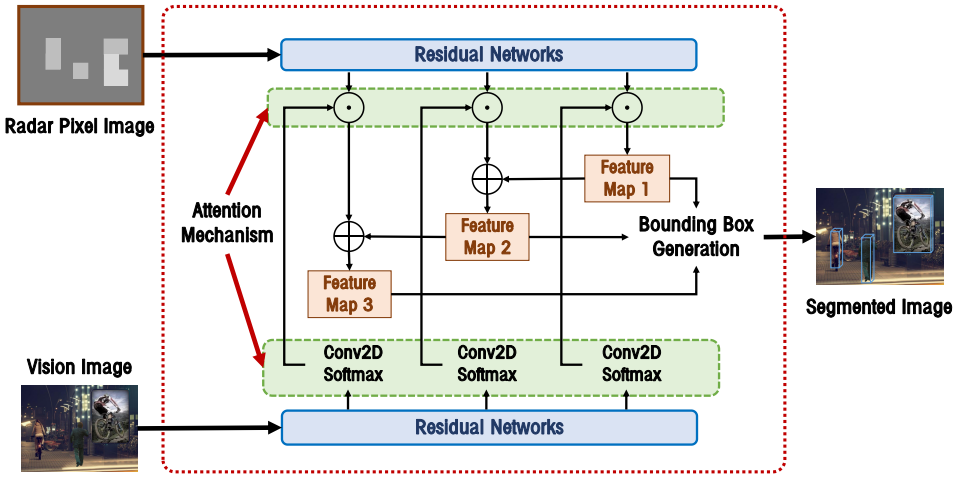


Fig. 8. Fusion Model for Target Segmentation

Because the mmWave radar and vision sensors are both on the autonomous vehicle and towards the same direction, the mmWave radar and vision data are regarded as different modalities of data of the same target from the same distance and angles of view. Therefore, this paper proposes the multi-modal features fusion methodology for associating those data in two different modalities, utilizing their relationship. Specifically, this paper utilizes the features extracted from the mmWave radar pixel data to detect the locations of the living target in the vision data. The segmentation and the detection process are similar to the process in the human brain while ignoring useless information and focusing on important information. The mmWave radar pixel data is regarded as the weight matrix for the segmentation and detection process. Technically, the segmentation and detection process is done by fusing the vision and the mmWave pixel data utilizing the attention mechanism [36]. The attention mechanism's multi-modal data fusion model is shown as follows:

$$\mathcal{F}_{i,j} = \varphi(\mathcal{V}_{i,j}) \odot \psi(\mathcal{R}_{i,j}) \tag{4}$$

Here the $\mathcal{F}_{i,j}$ is the fused data feature at the index (i, j), $\mathcal{R}$ is the feature from mmWave radar pixel data, $\mathcal{V}$ is the feature from vision data, the linear mapping operation is represented by φ, the two-dimensional convolution together with Softmax operation is represented by ψ, and $\odot$ represents for the Hadamada product operation. φ and ψ are the known parameters calculated from the sensors deployment setting.

The features in the vision and the mmWave radar pixel data are operated by the above-mentioned attention operation. Moreover, this article takes advantage of the feature pyramid network to provide extra context by feeding the features captured on the global scale to the local scale. Specifically, as shown in Fig. 8, the fusion data features in layer five are up-sampled and then concatenated with the fusion data features in layer four. Layer four contains a new up-sampling and concatenation algorithm for generating the data features for layer three. After that, the result of liveness target detection and distinguishment is generated in these three layers.

In this article, two fusion models are presented utilizing the idea of the feature pyramid network. As shown in Fig. 8, the first fusion model is designed based on radar pixel images and visual images for target segmentation in multi-target scenes. The fusion model puts the two image data into the residual network respectively and sets up an attention mechanism to improve the fusion performance. For the processing of visual images, this paper utilizes the Conv2D Softmax model to extract image details. Then, the information obtained from both sides is merged and discriminated at the pixel level, and the fusion results of different residual levels are utilized to generate features. Finally, the object's bounding box is generated based on these different levels of features.

In addition to realizing the object segmentation function of the image, this paper also implements the posture acquisition function through the posture detection method integrated with the MeidaPipe framework [3]. However, because this method is based on conventional image visual features, it cannot distinguish the difference between the living target and the metal from the fundamental principle, which leads to the wrong judgment of the human posture information on the metal. In order to solve this problem, this paper also proposes a fusion model based on posture information and object segmentation information obtained from the above-mentioned fusion model. As shown in Fig. 9, this model adopts a similar implementation mechanism as the previous fusion model, and the input data are posture information and target segmentation information, which are obtained from the above-mentioned target segmentation and posture detection process separately. Finally, several obtained bounding box schemes were screened and judged to achieve the correct bounding box of the living target.

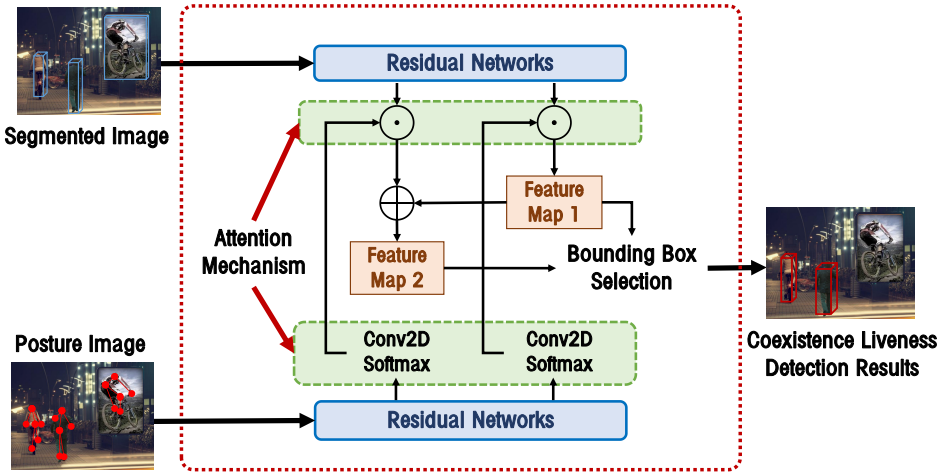


Fig. 9. Fusion Model for Liveness Target Detection

6.5 Liveness Target Detection Complexity

The pseudocode for our proposed liveness target detection algorithm is presented in Algorithm 1. The algorithm's time complexity primarily depends on three operations: prediction model execution, bounding box decoding, and Non-Max Suppression. The prediction phase has a time complexity of $O(n)$, with n being the total number of image pixels. Bounding box decoding, processed on each predicted bounding box, has a time complexity of $O(m)$, where m denotes the total number of predicted bounding boxes, i.e., the number of living targets. Following this, the Non-Max Suppression eliminates the inaccurate bounding boxes, characterized by a time complexity of $O(m^2)$. Therefore, the overall time complexity of the algorithm is $O(n + m^2)$. This balanced time complexity showcases an optimal trade-off between detection accuracy and computational efficiency, thereby meeting the crucial requirement of real-time response in autonomous driving systems.

In order to evaluate the latency of the proposed methodology in real-world scenario. We conduct target liveness detection on a laptop equipped with an NVIDIA GeForce RTX 4060 GPU and an AMC Ryzen 9 7845HX, which exhibits computational capabilities comparable to those of the hardware utilized in autonomous vehicles, such as the AMD Ryzen in the Tesla Model 3. The total system latency averages around 50ms. The processing of the mmWave radar signal and the extraction of the RCS pixel image contribute about 15ms to this latency. Meanwhile, the target liveness detection introduces approximately 35ms of latency. These metrics collectively attest to the system's viability for real-world applications.

Algorithm 1 Liveness Target Detection Algorithm

Input: *Images*, Camera Vision Images, *Radar*, Radar Pixel Image

Output: *Box*, Draw bounding boxes on the image

predict = **self.**net(*Image*, *Radar*)
output = **self.**bbox_util.decode_box(*predict*)
results = **self**.bbox_util.non_max_suppression(torch.cat(*output*, 1)
if *results* are empty **then**
  return *Images*
end if
top_label = np.array(results[0][:, 6], dtype = 'int32')
top_conf = results[0][:, 4] * results[0][:, 5]
top_boxes = results[0][:, :4]
% Draw bounding boxes on the image with *results*
return *Images* with bounding boxes

7 IMPLEMENTATION AND DATA COLLECTION

7.1 Data Collection

Data Collection Platform. Based on the existing commercial high-definition camera Logitech Pro C920 and commercial mmWave radar IWR6843, this paper designs and fabricates a mobile data collection platform to simulate an on-board system as shown in Fig.10. The configuration of the radar and camera is as follows: (1) The radar used in the experiment has three transmitting antennas and four receiving antennas. The operating frequency band is from 60GHz to 64GHz, the operating wavelength is ~ 4mm, the azimuth FoV is 60 degrees, the elevation FoV is 60 degrees, and the angular resolution is ~ 15°. The radar transmits 64 chirps per frame. The chirp starting frequency is 60GHz, the bandwidth is 1009.82MHz, and the frequency slope is 21.038MHz/us. (2) The camera's resolution used in the experiment is 1920×1080, and its frame rate is 30fps.

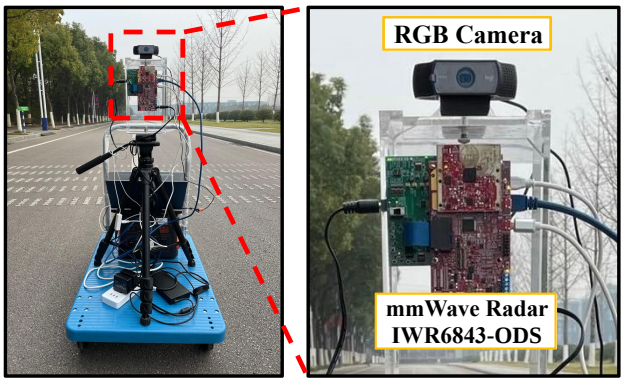


Fig. 10. Data collection platform.

Experiment Site. This paper conducts experiments using the data collection platform designed in the previous section. It simulates the scenario in question on an urban road in Hong Kong, where metals with images of hybrid target of

living entity and metal are placed on the side of the road, and real-live hybrid target of living entity and metal are also standing in the middle of the road. The platform is slowly moved from a distance to the billboard and the hybrid target of living entity and metal, and the distance is about 30 meters. While moving, two devices (a millimeter wave radar and a camera) collect data simultaneously. This paper conducts eight sets of experiments, collecting a total of 26050 frames of image data and 5210 frames of radar data. In the process of training the model, the ratio of the validation set to the test set of the training data is set to 9:1. Finally, to avoid the situation of training overfitting caused by the image unchanged in the metal, this paper replaces the metal rider pattern in the data with other rider images that do not exist in the data set.

7.2 Object Segmentation for Radar Data

The problem of noise generated by multipath effects is difficult to solve for all RF techniques [15]. In this paper, the main scene is outdoors, in which, in addition to the identified targets (portrait metal and hybrid target of living entity and metal), there are also some other static obstacles (trees and ground). The reflection and beam spread generated by these surrounding objects will lead to the multipath effect and eventually lead to a large number of interference information called ghost points in the point cloud information. This paper uses the point segmentation strategy based on the DBSCAN clustering method to solve these ghost points and implement image segmentation. DBSCAN is a density-based clustering algorithm based on space distance and domain density to achieve point cloud division, so it is not sensitive to noise. Therefore, it alleviates the multipath effect of noise information on the experimental image. DBSCAN does not need to set the number of clusters in advance and automatically mark noise points and outliers, so it is often used to separate objects in processing millimeter wave radar point cloud data. Based on experience, in this paper, the maximum distance (radius) between two points falling into the same cluster is set to one, and the minimum number of points within a cluster is set to three.

8 EVALUATION

8.1 Evaluation Methodology

Evaluation Metrics. In this paper, we choose four parameters, Precision, Recall, F1 Score, and mean Average Precision (mAP), as evaluation indicators to measure the effect of the target detection task. Accuracy is equal to the number of true positive samples predicted as positive samples divided by the number of true positive samples. Recall equals the number of correctly predicted positive samples divided by the number of all positive samples. The F1 score is a comprehensive consideration of the precision and recall of the classification model. It is an indicator utilized to measure the classification accuracy of the binary classification model, which is regarded as the harmonic mean of the model's precision and recall. In this paper, IOU is set as the intersection and union ratio of the sample 3D bounding box and the real value 3D bounding box and the true and false attribute values of the sample 3D bounding box are determined based on the intersection and union ratio of the sample 3D bounding box and the real value 3D bounding box. When the IOU of the sample 3D bounding box and the real value 3D bounding box are greater than the set threshold, the sample is judged to be true.

Average Precision (AP) is equal to the area under the precision and recall curves determined by the confidence threshold, so it is utilized to show the overall performance of detection methods under different confidence thresholds. To obtain AP, the first step is to utilize the trained model to find a confidence score for all the bounding boxes and rank them based on that score. The second step is to select the top-1 to top-n results and compute precision and recall for the

number of boxes. The last step is to plot all the precision and recall to form the P-R curve and then find the area under the P-R curve is the AP value. The mean average precision (mAP) is the average AP of each class.

Competing Approaches.

- **Pure RCS:** This approach provides a fundamental insight into the potential of utilizing RCS alone for liveness target detection. The pure RCS approach does not take into account the multi-faceted influences of an object's orientation, shape, and viewing angle on the RCS values. Therefore, the accuracy is not good enough. However, the limitations of RCS are able to be mitigated by collaborating it with vision data. This fusion leverages the strengths of both modalities, creating a more comprehensive and reliable detection mechanism.
- **milliEye [34]:** Xian et al. proposed the milliEye in 2021. milliEye achieves target detection in low-light environments utilizing the vision and mmWave radar fusion. milliEye exploits the millimeter-wave radar's insensitivity to light and combines 3D radar point clouds with visual images for liveness target detection. It is considered a two-stage detector and mmWave radar point cloud fusion method without RCS information. In this paper, milliEye is applied to the dataset to evaluate its performance.
- **Faster-RCNN and mmWave Radar RCS (Faster+RCS):** This is a two-stage detector and radar point clouds fusion method with RCS information, which is applied to the dataset in this paper for calculating the liveness target detection performance.
- **w/o attention mechanism (No-Attention):** This paper replaces the proposed attention mechanism-based feature fusion method with the tensor concatenation, which aims to investigate the influence of attention mechanism on performance.
- **w/o RCS (No-RCS):** This method replaces the RCS features in the mmWave radar data with the received signal intensity and makes a new radar pixel image dataset for liveness target detection. This experiment group aims to verify the effectiveness of RCS in liveness target detection.

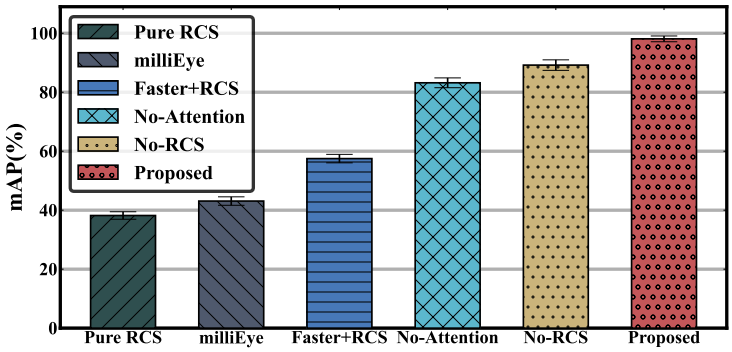
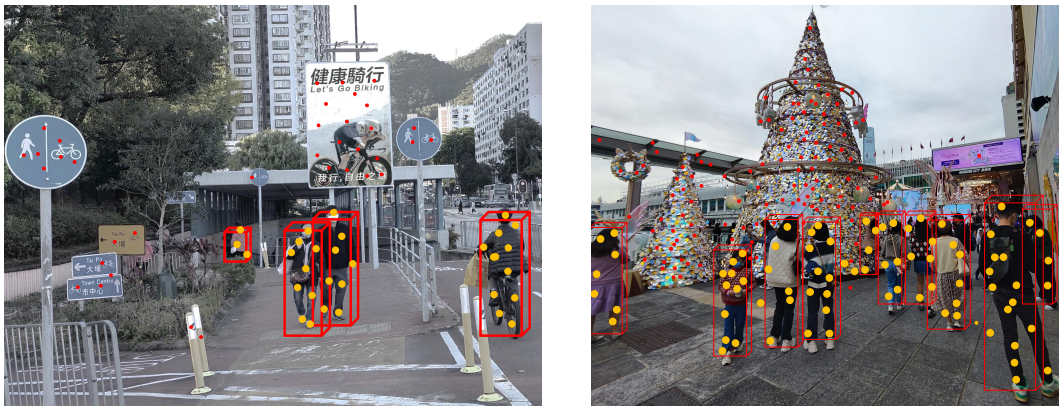


Fig. 11. Overall performance.

8.2 Overall Performance

The proposed approach and four other baseline methods are compared, and a picture of the overall performance is plotted as shown in Fig.11. It is seen from the figure that the mAP of the proposed method reaches 98.1%, which is higher than all the other four baseline methods. The mAP of the proposed approach is 8.3% higher than that of the **No-RCS** method and is 14.9% higher than the **No-Attention** method. It shows that the radar's RCS value is better than its absolute intensity value in distinguishing between the living target and visual interference. The attention mechanism also helps to improve the fusion effect. Finally, both **Faster+RCS** and **milliEye** have low mAP, which indicates that

there are obvious problems with the current two-stage method for detecting living targets in autonomous driving scenarios, such as inaccurate detection results of small-scale and long-range activity. Fig.12 shows two examples of liveness target detection in this paper, where yellow points indicate the point cloud of live targets with low RCS values, and red points indicate the point cloud of the billboard with high RCS values. Fig. 12(a) demonstrates the liveness target detection on an open road, with bikers and pedestrians as the living targets, and a billboard with a biker figure as the interference. The result indicates that living targets are detected accurately. Furthermore, to test the maximum number of detectable targets, we evaluate our method in a crowded environment. Fig. 12(b) illustrates the liveness target detection at a shopping plaza, with pedestrians as the living targets, and a metal Christmas tree as the interference. The result indicates that our method performs accurate liveness target detection for more than ten targets simultaneously in a crowded environment.



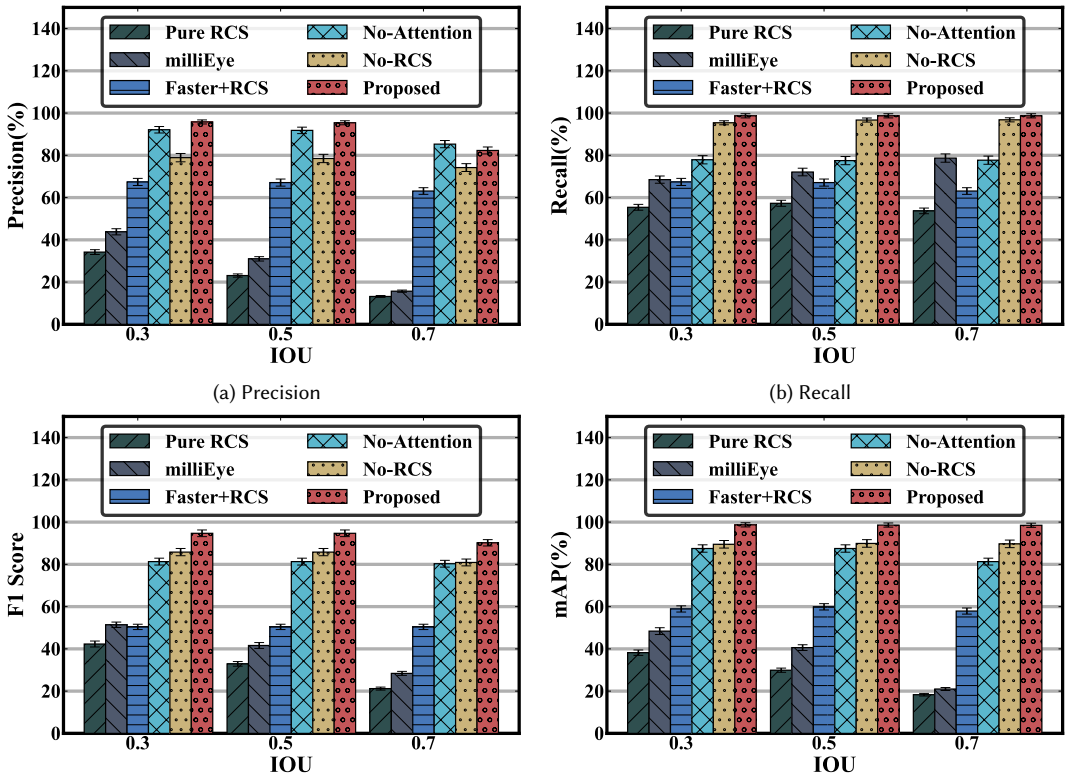
(a) Example on an open road. (b) Example at a shopping plaza.

Fig. 12. Examples of liveness target detection.

8.3 Sensitivity Analysis

Precision, Recall, F1 Score, and mAP. Four evaluation indexes, namely precision, recall, F1 score, and mAP, are utilized to measure the performance of the proposed method. As shown in Fig.13, the proposed method is compared with five baseline methods. As a whole, the proposed method outperforms all the baseline methods in every metric. Specifically, regarding the **Pure RCS** method, it provides a fundamental perspective on utilizing RCS alone for liveness target detection. However, its performance is less satisfactory due to its failure to consider factors such as an object's orientation, shape, and viewing angle on the RCS values, which are accounted for in our proposed method. Furthermore, as illustrated in Fig.13(a), the accuracy of the proposed method, denoted as **Proposed**, exceeds that of the **NO-RCS** method. This is attributable to the replacement of radar signal strength with RCS value, which allows for a more effective distinction and measurement of living target characteristics. As shown in Fig.13(b), the recall rate of **Faster+RCS** method is relatively low because this method cannot accurately carry out the target teaching material and segmentation. Fig.13(c) illustrates that the overall performance of **milliEye** method is significantly lower than other methods due to the lack of multi-modal fusion strategy and the RCS of mmWave radar, which also confirms that the method based on computer vision is susceptible to the interference of the external environment.

Impact of IoU threshold. The IoU threshold of the predicted active target and the real active target is set in the evaluation process, and the performance of each method in the case of different IoU values is evaluated. When the IoU



(a) Precision (b) Recall

(c) F1 Score (d) mAP

Fig. 13. Performance of proposed and baselines in different IoU thresholds

threshold is larger, the bounding boxes of small-scale targets are more likely to be lost. As shown in Fig.13(d), the mAP of **Proposed** always remains above 97% and is better than the other four baseline methods, which indicates that the proposed method has high robustness. In addition, it is seen from the data in the figure that the performance of the method based on the single-stage multi-scale detector is significantly higher than that based on the two-stage detector, which proves that the multi-scale method realize the detection of small-scale targets in long-distance scenes.

Impact of target distance. In this paper, the performance results for different distance cases are collected, and the amount of data to keep different distance cases is the same. As shown in Fig.14, although the distance between the target and the experimental platform will affect the proportion of the target imaging in the two kinds of sensing devices, the mAP of the proposed method is stable at 97% in the experimental scenarios with different numbers of living targets and different distances. Importantly, beyond a 30-meter range, the detection performance of the commercial mmWave radar dramatically declines. This 30-meter range is sufficient for an autonomous vehicle to perform essential braking or evasive maneuvers [22], setting the optimal functioning range of the system. Despite this constraint, our method demonstrates significant robustness to images captured at varying distances within the commercial mmWave radar's detection scope.

Impact of the number of living targets number. This paper also collects the experimental results under different living target numbers, divides the data set according to the different living target numbers, and evaluates the performance

of the proposed method under these conditions. As shown in Fig.14, when the distance between the experimental target and the experimental platform is maintained to a certain extent, the overall mAP of the proposed method always remains above 97%, which indicates that the proposed method is robust to changes in the number of living targets. Because when the target gets closer to the mmWave radar, the display of the target on the mmWave radar is incomplete, and the performance increases when the target distance increases.

Moreover, we also evaluate the performance of liveness target detection under different target angle-of-view. To show the impact of the angle-of-view more clearly, we select the three most representative target angle-of-view as an example. As shown in Fig. 15, because the side view of the hybrid target of living entity and metal has more information than the front, the proposed model has higher performance when facing the side view of the hybrid target of living entity and metal than the front view of hybrid target of living entity and metal. The living target has less detection accuracy because it has less RCS and information.

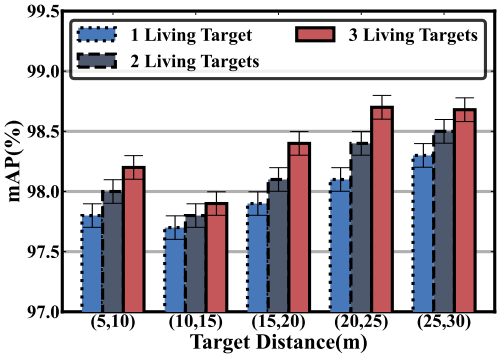


Fig. 14. Performance with target distance.

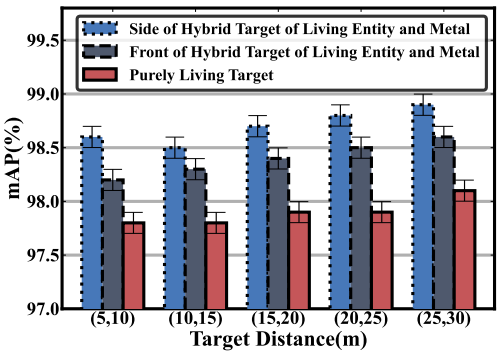


Fig. 15. Performance with target angle-of-view.

Impact of the Angle of View. Fig.16 illustrates the performance variation across different angles of view. Two distinct target scenarios are considered in the experiment: a hybrid target of living entity and metal, and a target representing a purely living target. The observation angles, extending from 0° to 180° at 30° intervals, distinctly affect the detection performance for both scenarios. At the observation angles of 0° and 180°, there is a lesser amount of data available, leading to a minor decrease in performance. Conversely, the accuracy exhibits a proportional increase with the observation angle, culminating at a peak at 90°. This peak is attributable to the optimal target visibility at this angle, which maximizes data capture for the millimeter-wave radar and camera systems, thereby optimizing the mean Average Precision (mAP). Importantly, despite these variations, the mAP consistently remains above the 97% benchmark throughout our experiments, underlining the system's robustness and reliability across different viewing angles.

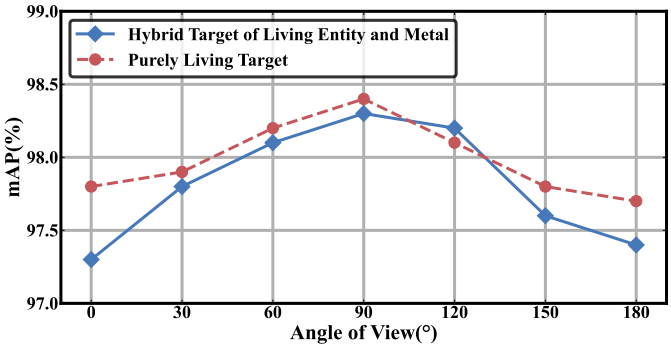


Fig. 16. Performance with different angle of view

Impact of different road scenes. In the actual road scene of vehicle driving, the radar data will be significantly affected by the road conditions, the distribution of trees, and street lights on both sides of the road. As shown in Table.2, the detailed road environment information is collected in this paper.

In this paper, different numbers of living targets are experimentally evaluated in four road scenarios. The results show that the mAP of the proposed method is 97.4%, 97.3%, 97.1%, and 97.7% in four scenarios, respectively. The experimental results show that the proposed method based on RCS is adaptive to the change of road scene and environmental heterogeneity because the influence of environmental noise on radar SNR is small, and the preprocessing algorithm proposed in this paper removes static reflection points in different environments.

Table 2. Differences of Road Scenes

Scene	Condition	Trees Location	Lamps	Existing Living Target
1	Rough	Road End	Not exist	1
2	Rough	Not exist	Not exist	2
3	Flat	Both sides	Exist	1 & 2
4	Flat	Road End	Exist	2 & 3

Impact of sunlight condition. Fig. 17 demonstrates the efficacy of the proposed target liveness detection method functioning across a spectrum of illumination conditions. It emphasizes the mean average precision (mAP) as a metric for identifying one to three live targets. As the ambient illumination transitions from low-light to highly-lit environments, a corresponding increase in the system's precision is noted. This is exemplified when the light intensity escalates from a minimal 0.1 lux to a significantly brighter 10^5 lux, catalyzing an enhancement in the mAP from 97.1% to 98.7%. This progression is attributed to abundant visual information available under high-illumination scenarios. However, over-intense lighting induces interference, primarily due to the mirror-reflection effect. This interference becomes evident when the light intensity exceeds 10^5 lux, leading to a drop in the mAP to 97.7%. Despite these variations in illumination, the proposed algorithm exhibits remarkable robustness, consistently maintaining an accuracy level above 97%. This underlines the liveness target detection algorithm's ability to uphold high precision across a wide spectrum of lighting conditions.

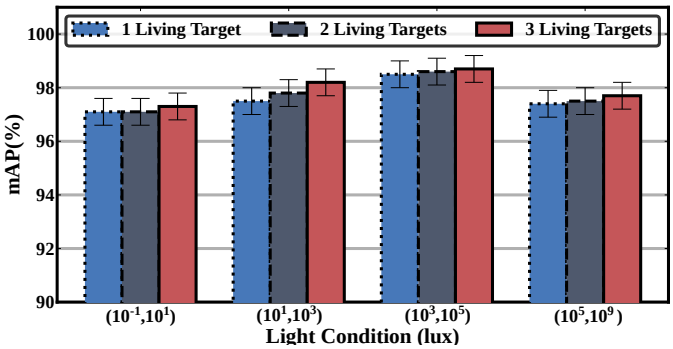


Fig. 17. Light Condition.

In addition, to better evaluate the model's generalization ability, this paper also conducts an experimental evaluation on scenes with four different lighting conditions, shown in Table.3. For example, in scenes 1&2, the camera will be affected by the direct or indirect reflection of sunlight from the target, and the target detection accuracy is low. In scene 4, the blur of the target image caused by weak light also affects the effect of target detection. However, the above

experiments show that the mAP of the proposed method reaches more than 97% in all four scenes, which indicates that the proposed method is robust to different light conditions. The result also verifies that millimeter wave radar is not sensitive to ambient weather.

Table 3. Differences of Sunlight Conditions

Scene	Experimental Condition	Targets' Lighting Condition
1	Face the sun, strong light	Back to the sun ,shadow
2	Back to the sun, shadow	Face the sun, strong light
3	Cloudy	Ordinary light brightness
4	Nightfall	Low light brightness

Impact of occlusion condition. The occlusion of both living targets and non-living targets influences detection accuracy. In cases of occlusion among living targets, an increase in the target count inevitably leads to overlaps within the sensing scope, consequently impacting the operational limits of the system. This scenario commonly arises when the radar's sensing range is simultaneously occupied by over 20 human targets, resulting in an occlusion rate that exceeds 60%. In terms of occlusion between living targets and other objects, mmWave radar signals demonstrate the ability to penetrate certain types of occlusions, such as wood, but are virtually incapable of penetrating metallic occlusions. To evaluate the system under stringent conditions while maintaining experimental uniformity, we introduce metallic occlusions during occlusion experiments. A metal plate serves as the occluder to obscure the liveness target at varying occlusion rates. Fig. 18 presents the mean Average Precision (mAP) accuracy for the detection of 1, 2, and 3 living targets amidst varying levels of occlusion. An escalation in occlusion rate results in a diminished quantity of information from both mmWave radar and vision data, thereby causing a decrease in mAP. As occlusion rates range from 0% to 100%, a concomitant decline in mAP scores manifests, highlighting the adverse effect of occlusion on target detection.

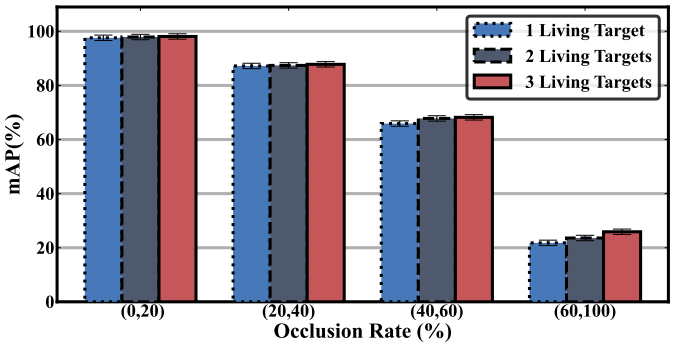


Fig. 18. Occlusion Rate.

9 DISCUSSION

In 2020, 30.6% of fatal crashes were caused by cars colliding with moving objects, e.g., animals, pedestrians, and bikers [29]. The end-to-end liveness target detection methodology proposed in this article is crucial for improving the robustness of autonomous vehicles to prevent accidents caused by visual interference. Instead of proposed pedestrians and bikers, animals and other kinds of liveness have similar RCS features in the mmWave radar data. Therefore, the proposed methodology is suitable for detecting different kinds of living targets and improving the safety of autonomous vehicles.

10 RELATED WORK

Table 4 provides an overview of the current state of target object detection methodologies, categorizing them into two main streams: Single-modal and Multi-modal approaches. The majority of existing techniques that utilize Radar Cross Section (RCS) are predominantly single-modal, focusing primarily on target detection. However, these approaches do not extend to the more intricate task of target liveness detection. In contrast, while multi-modal techniques show promise in providing a more holistic view of the scene, they typically do not incorporate RCS and also fall short in addressing target liveness detection in the proposed scenario. Significantly, our work stands out as the first to introduce a multi-modal target liveness detection method that incorporates RCS. This unique contribution bridges the gap in current methodologies, expanding the application of RCS from Single-modal to Multi-modal detection, and extending the scope of detection from mere target identification to the more advanced task of target liveness detection.

Table 4. Related Target Detection Approaches

	RCS-Based	**No RCS-Based**
Single-model	[9, 11, 13, 16, 33]	[5, 8, 14, 24–27, 30]
Multi-modal	This Work	[12, 19, 21, 32, 38, 40]

10.1 Single-modal Detection

Because the road environment is often very complex in the actual driving scene, achieving high-quality liveness target detection is crucial to the autonomous driving technology of vehicles. Therefore, there has been a lot of research and progress in this regard, and the technology of using various sensors to achieve object detection is relatively very mature. In the aspect of computer vision, there are two commonly used methods to realize liveness target detection. One is based on the contour feature of the face to realize liveness target detection [26], but this method is only used in the case of close faces, so it is not suitable for automatic driving scenes. Another type of method is based on the contour features of the human body to realize the detection of living targets [25, 27], and the implementation effect of this method in the autonomous driving scene is relatively much better. However, these methods do not consider the characteristics of living factors, so although they distinguish people from other non-human objects, they will produce wrong judgments when distinguishing between real people and dummies on metals. On the other hand, because millimeter wave radar is not sensitive to the environment, there are also many related studies to detect living targets for vehicular millimeter wave radar. For example, in reference [5, 24], the heat map information of the radar point cloud is input into the neural network to classify different types of objects on the road. In reference[14], the radar Doppler spectrum was used as the motion feature to realize the distinction between vehicles and living targets. In literature [8, 30], radar point clouds are used for road target classification and vehicle detection.

RCS, a fundamental parameter within the domain of mmWave radar, encapsulates the inherent reflective properties of an object when exposed to radar waves, thus facilitating its application in target detection. The capacity to measure the intensity of the signal reflected by a target, along with the consistent persistence of this reflected intensity for a specific object, emphasizes its reliability. Furthermore, the reflective phenomena of RCS show considerable differences between living and non-living targets, suggesting its potential applicability in liveness target detection. Nevertheless, despite these promising attributes, to the best of our knowledge, there exists no research that leverages RCS for liveness target detection. Existing studies primarily concentrate on utilizing RCS for distinct objectives. For instance, Fu et al. [11] propose an identification and classification technique for unique drone types based on their RCS signatures. They

offer a deep learning-based methodology capable of differentiating between various drone models and orientations utilizing a CNN trained on simulated RCS data. Furthermore, Shibao et al. [33] analyze the RCS patterns of diverse types of road debris and put forward a debris detection algorithm that reduces false alarms induced by clutter. However, the utilize of RCS for distinguishing living targets remains uncharted territory. This gap in the current literature highlights the novelty and significance of our research. In this paper, we pioneer a novel application of RCS for liveness target detection.

10.2 Multi-modal Detection

Although the research on target detection based on single mode has made much progress, the method based on vision is easily affected by the environment, weather, and object material, and the method based on millimeter wave radar is limited by the sparse and noise characteristics of radar. Therefore, using multi-modal fusion methods to make up for the shortcomings of different sensors has become the mainstream strategy in the field of autonomous driving, including the fusion of millimeter wave radar and camera. Among them, there are two types of fusion of millimeter wave radar and visual image. One is to utilize millimeter wave radar point cloud [21, 32, 40] to fuse with the visual image for target detection, and the other is to utilize Doppler heat map of millimeter wave radar [19, 38] to fuse with the visual image for target detection. However, these two types of methods do not consider the detection characteristics of liveness. Although they have achieved good results in object detection, they often have false recognition when distinguishing liveness and virtual images. In addition, 3DCNN[12] is the only work for liveness target detection at present, but 3DCNN is a method based on lidar point cloud and visual image fusion, and there is still a problem that it cannot distinguish between real and fake when facing virtual human images such as metals.

In order to solve the problem of liveness target detection in unmanned driving scenarios, this work proposes a detection method using millimeter-wave radar and vision fusion, and for the first time, utilizes radar cross section (RCS) as a recognizable feature for liveness target detection. Our experimental results also show that there is a vast difference between the RCS of an active target and its visual interference, such as an LED billboard, due to material, pose, and other differences in the reflection properties. In addition, the RCS values are stable in the detection range of the mmWave radar, which means that the results are not affected by the distance from the target to the radar.

11 CONCLUSION

This paper presents an end-to-end liveness target detection methodology with mmWave radar and vision fusion. The proposed method improves the robustness of autonomous vehicles while facing visual interference. The RCS feature extracted from the mmWave radar data is the crucial distinguishing characteristic of the liveness target and the visual interference. We propose a multi-modal feature extraction method to extract features from the mmWave radar and the vision data. Then, we propose multi-modal feature fusion methodologies to fuse these features to distinguish living targets from visual interference. The proposed system in this article improves the robustness of autonomous vehicles by enabling the capabilities of precise liveness target detection.

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